

mixture interpolation

locations

masses

Displacement vs Mixture Interpolation

The simplest alternative to McCann is the ‘naive’ **mixture interpolation**

Displacement Interpolation:

$$\rho_t^{\text{disp}} = [(1 - t)\text{Id} + tT]_{\#}\alpha$$

Interpolates **locations** of points

Mixture Interpolation:

$$\rho_t^{\text{mix}} = (1 - t)\alpha + t\beta$$

Interpolates **masses** of points

Mixture interpolation is easy to compute, but not* a geodesic in $\mathbb{W}_p$

Recall from last time:

- Warm-up: simple dataset with structure
- First attempt: “normal” (euclidean) averaging
- This **ignores geometry!**
- Is there a better way?
- OT can average in a meaningful way

